Experiment : 5

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Subject Name: Full Stack Development – II **Subject Code: 23CSH-309**

1. **Aim**: To verify the correctness and reliability of the EcoTrack React application by writing automated tests using Jest and React Testing Library, and by analyzing application behavior using debugging tools.

2. **Objective**:

• Understand the purpose of automated testing in frontend applications
• Write unit tests for JavaScript utility functions using Jest
• Use different Jest matchers to validate expected outputs and behaviors
• Test React components using React Testing Library
• Verify UI rendering by querying elements from the DOM
• Implement asynchronous testing using findBy and waitFor methods
• Apply mocking to simulate API or external data responses in tests
• Perform snapshot testing to detect unintended UI changes
• Debug failing tests and application logic using browser Developer Tools and breakpoints
• Analyze application behavior and errors systematically rather than manual checking

3. **Implementation / Code:**

- **Tools & Technologies Used:-**

• React.js
• JavaScript (ES6)
• Jest Testing Framework
• React Testing Library
• VS Code
• Node.js & npm
• Web Browser (Chrome DevTools)

- **Implementation Description:-**

• The EcoTrack application is tested to ensure correctness of both logic and UI behavior.
• Unit testing is performed on utility functions (e.g., calculator function) using Jest.
• React Testing Library is used to render components and verify UI structure.
• Snapshot testing is applied to detect unintended UI changes over time.

• Automated tests improve application reliability and maintainability.
• Debugging tools such as browser DevTools and breakpoints help identify errors in logic or rendering.

- **Sample Code Snippet:-**

```
const reportWebVitals = onPerfEntry => {
  if (onPerfEntry && onPerfEntry instanceof Function) {
    import('web-vitals').then(({ getCLS, getFID, getFCP, getLCP, getTTFB }) => {
      getCLS(onPerfEntry);
      getFID(onPerfEntry);
      getFCP(onPerfEntry);
      getLCP(onPerfEntry);
      getTTFB(onPerfEntry);
    });
  }
};

export default reportWebVitals;
```

JS Tracker.test.js ×

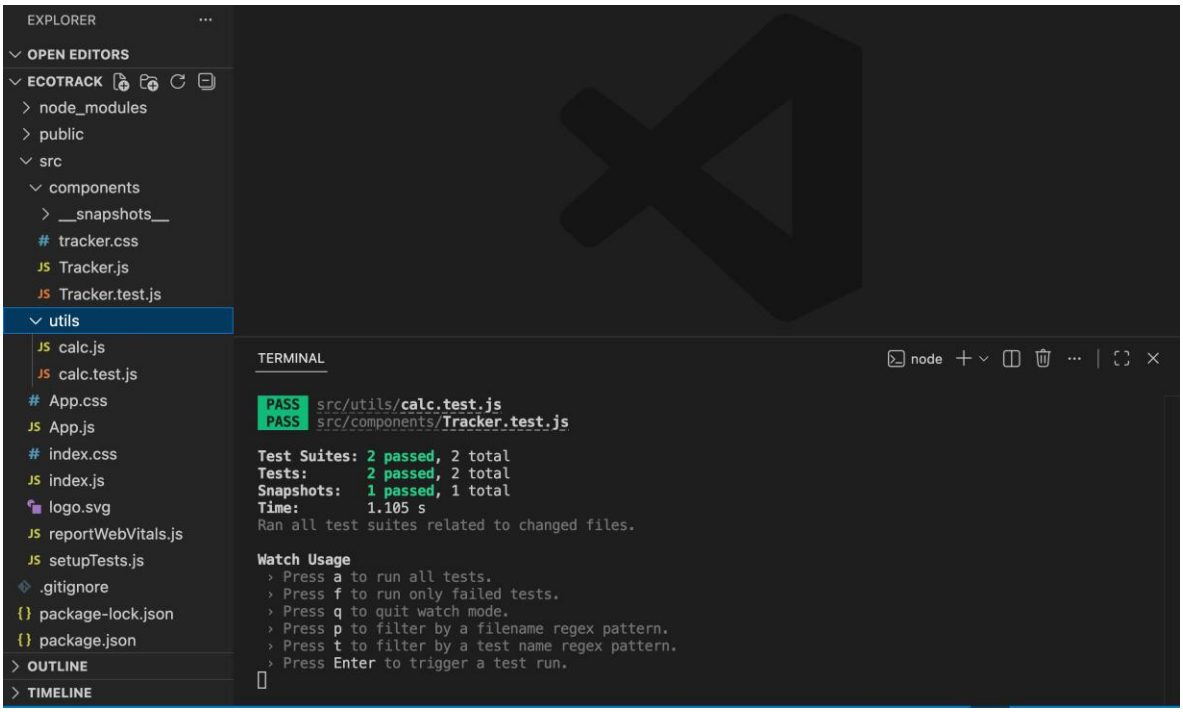
src > components > JS Tracker.test.js > ...

```
// import { render, screen } from "@testing-library/react";
// import Tracker from "./Tracker";

// test("loads async data", async () => {
//   render(<Tracker />);

//   const text = await screen.findByText(/Eco data loaded/i, {}, { timeout: 3000 });

//   expect(text).toBeInTheDocument();
// });

import { render } from "@testing-library/react";
import Tracker from "./Tracker";

test("matches snapshot", () => {
  const { asFragment } = render(<Tracker />);
  expect(asFragment()).toMatchSnapshot();
});
```

```javascript
import { add, getUserName } from "./math.js";

test("add numbers:", () => {
    expect(add(1, 2)).toBe(3);
});
```

4. Output:

- All Jest test cases executed successfully
- Utility function test passed
- React component snapshot test passed
- No unintended UI changes detected
- EcoTrack component rendered correctly during testing
- Debugging tools confirmed correct state updates and DOM rendering



5. Learning Outcomes (What I Have Learnt):

- Importance of automated testing in frontend applications
- Writing unit tests using Jest framework
- Using matchers like toBe() and toMatchSnapshot()
- Testing React components with React Testing Library
- Validating UI rendering through DOM queries
- Understanding snapshot testing for UI stability
- Debugging React applications using DevTools and breakpoints
- Improving software reliability and maintainability through testing